Where Geometric Brownian Motion Breaks Down: Fat Tails, Volatility Clustering, and Persistent Volatility Shocks in the S&P 500

L. D. Visconte

October 2026

Abstract

Geometric Brownian Motion (GBM) remains the baseline model for asset prices, yet its assumptions of normal, independent, constant-volatility returns are rarely accurate descriptions of market behavior. Using 2,519 daily S&P 500 log returns(September 2016–September 2026), this paper shows that GBM displays fat-tailing (kurtosis 19.7), volatility clusters, and volatility shocks persist for roughly 100 trading days. Four extensions (Merton jump diffusion, Variance Gamma, Heston and, Bates) are then tested on a common scale using probability integral transform (PIT) residuals. It is made clear then that Jump Diffusion repairs the shape of returns while stochastic volatility repairs most of the dynamics. Finally, this paper proposes a non-linear barrier jump model in which jump risk rises as prices approach an economic barrier (B). It achieves the highest likelihood and best AIC of all models tested, although BIC still favors Heston due to over-parameterization concerns.

1 Introduction

For centuries, random noise, random walks, and what seems like abstract movements have caused numerous headaches for mathematicians, scientists, and financial traders alike. Modeling seemingly random fluctuations, whether in a particle suspended in a fluid or an asset price, can be extremely difficult. This inability to model asset price movements specifically perplexed the field of quantitative finance for years, until Louis Bachelier, in his *Theory of Speculation*, suggested using Brownian motion to model financial options pricing [Bac00]. This idea paved the way for quantitative traders to model asset price paths using continuous-time stochastic processes, or random walks.

However, Bachelier's thesis was not without its flaws. Standard Brownian Motion has two main problems when used to model asset prices:

1. It can drop below zero, whereas asset and commodity prices cannot trade in negative values.
2. It utilizes absolute changes in position, which poses a practical issue in financial applications: a \$2 increase in a \$1 stock represents a drastically different percentage change than a \$2 increase in a \$500 stock.

In 1959, M.F.M. Osborne developed Geometric Brownian Motion (GBM), which uses logarithmic changes in price to eliminate the limitations of Standard Brownian Motion [Osb59]. Geometric Brownian Motion became the gold standard for modeling financial transactions when Black, Scholes, and Merton (1973) used it to derive their famous option pricing formula, kickstarting modern quantitative finance as we know it today [BS73].[1]

Economists were drawn to this framework because it rendered the math behind market fluctuations neat, predictable, and most importantly, solvable in closed form. Before exploring further, it is essential to define how Standard Brownian Motion (SBM) and Geometric Brownian Motion (GBM) operate, where they overlap, where SBM fails, and where GBM may fail as well.

[1] Merton's contemporaneous derivation of the same formula is [Mer73].

1.1 Standard Brownian Motion (The Wiener Process)

Standard Brownian Motion is formally defined as the Wiener Process, denoted W_t. It is a continuous-time stochastic process used to model pure random noise, serving as the mathematical baseline for modeling unpredictable, continuous paths over time. The Wiener process satisfies the following fundamental properties:

1. **Initial State:** Starts at zero ($W_0 = 0$).
2. **Independent Increments:** The change in value over any time interval is completely independent of changes over past, non-overlapping intervals.
3. **Stationary Normal Increments:** Over any time step $t - s$, the change $(W_t - W_s)$ follows a normal (Gaussian) distribution with a mean of 0 and a variance equal to the elapsed time step $(t - s)$, such that $(W_t - W_s) \sim \mathcal{N}(0, t - s)$.
4. **Continuous but Non-Differentiable Paths:** The path is continuous across time, but because its direction changes instantaneously at every point, it is nowhere differentiable, meaning its instantaneous velocity over time is undefined.

Geometric Brownian Motion (GBM) Geometric Brownian Motion is a continuous-time stochastic process where the logarithm of a variable follows a Standard Brownian Motion with drift. It is defined by the following Stochastic Differential Equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{1}$$

where S_t represents the asset price, μ is the drift or expected growth rate, σ is the constant volatility, and dW_t is a Standard Brownian Motion increment.

Applying Itô's Lemma yields the explicit closed-form solution:

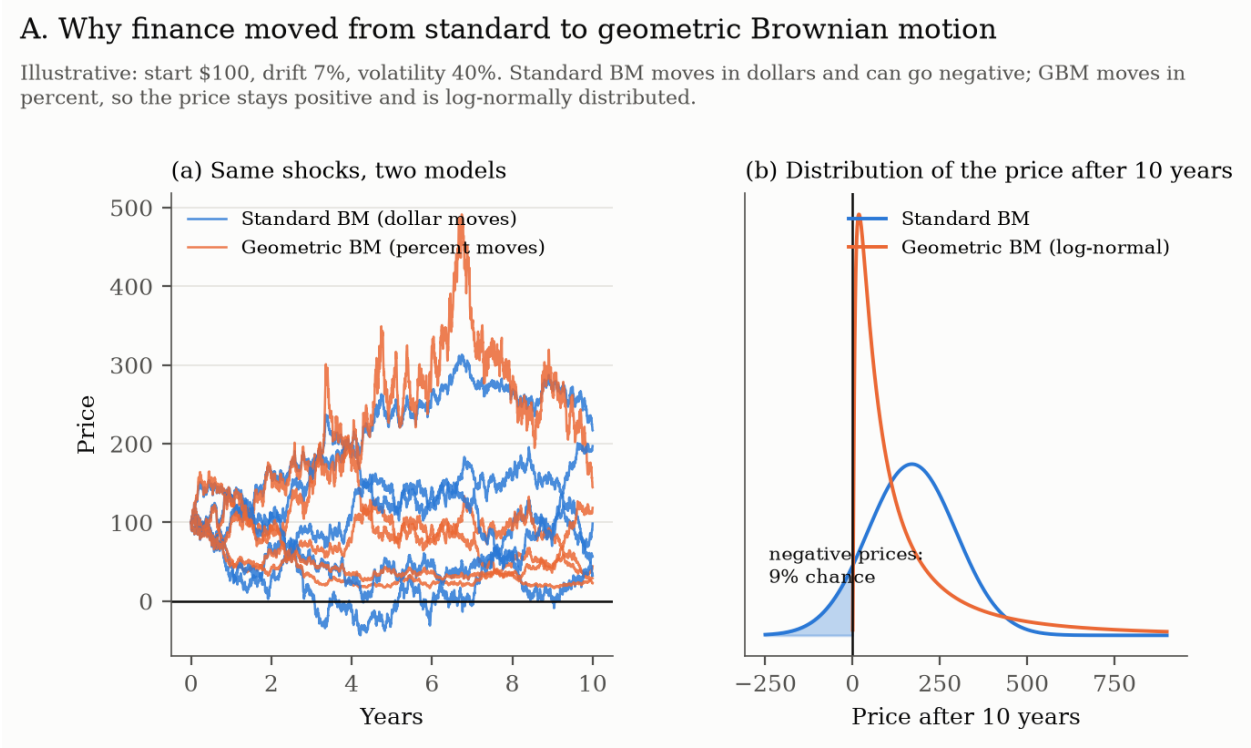
$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right) \tag{2}$$

Under this formulation, the logarithmic returns $\ln(S_t/S_0)$ are normally distributed, which means that the price of the asset S_t itself follows a logarithmic normal distribution. Furthermore, because S_t is an exponential function, the asset price is strictly positive ($S_t > 0$).

Problem Statement & Paper Structure: Although GBM is mathematically tractable, real financial markets are complex and often driven by uncertainty and sudden events. Therefore, this paper explores the following question:

> *To what extent does Geometric Brownian Motion accurately reflect real-world asset pricing dynamics, and where do its core mathematical assumptions fail when tested against empirical financial market data? What alternatives/additions to Geometric Brownian Motion have been adopted, and where do their core assumptions fail when posed against realized market data? And is there a synthesis of alternatives/additions that can be creatively manipulated to more accurately model asset pricing?*

The remainder of this paper is organized as follows: Section 2 details the derivation of the SDE using Itô's Lemma; Section 3 outlines the statistical testing methodology; Section 4 analyzes empirical failures of Geometric Brownian Motion; Section 5 discusses alternatives/additions to Geometric Brownian Motion; Section 6 will rigorously test those alternatives/additions; Section 7 provides a unique alternative that serves to patch one some of the faults with other modeling methods; Section 8 empirically tests that unique solution and analyze the results; Section 9 serves to conclude the paper.



2 Derivation of SDE using Itô's Lemma

We begin with the stochastic model for Geometric Brownian Motion via its Stochastic Differential Equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{3}$$

We then transform the dS_t so that $f(S_t) = \ln(S_t)$ when $S_t > 0$.
We follow by taking 1st and 2nd order derivatives of the $f(S_t)$ giving us:

$$f'(S_t) = \frac{1}{S_t}, \quad f''(S_t) = -\frac{1}{S_t^2} \tag{4}$$

We now must find the $(dS_t)^2$ term in order to have all the inputs for Itô's Lemma, which reads as the following (noting the partial derivative of f with respect to t moves to zero):

$$d(\ln S_t) = f'(S_t)dS_t + \frac{1}{2}f''(S_t)(dS_t)^2 \tag{5}$$

The $(dS_t)^2$ term (using the dS_t equation model):

$$(dS_t)^2 = (\mu S_t dt + \sigma S_t dW_t)^2 \tag{6}$$

Expanded:

$$(dS_t)^2 = \mu^2 S_t^2 dt^2 + \sigma^2 S_t^2 dW_t^2 + 2\mu\sigma S_t^2 dt dW_t \tag{7}$$

We know dt^2 drops to zero, and dW_t times dt also tends to zero, along with the fact $(dW_t)^2 = dt$; these are known as Itô's Rules. With them in mind the equation looks as such now:

$$(dS_t)^2 = 0 + 0 + \sigma^2 S_t^2 dt \tag{8}$$

Now we can plug all these inputs back into Itô's Lemma:

$$d(\ln S_t) = \frac{1}{S_t}(\mu S_t dt + \sigma S_t dW_t) + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(\sigma^2 S_t^2 dt) \tag{9}$$

Simplified:

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t \tag{10}$$

And now we can integrate this equation from 0 to time t (substituting t for u for integration purposes):

$$\int_0^t d(\ln S_u) = \int_0^t \left(\mu - \frac{1}{2}\sigma^2\right) du + \int_0^t \sigma dW_u \tag{11}$$

The first integral can be found using the Fundamental Theorem of Calculus $\left[\int_a^b f(x) = F(b) - F(a)\right]$. Doing so yields:

$$\int_0^t d(\ln S_u) = \ln S_t - \ln S_0 = \ln\left(\frac{S_t}{S_0}\right) \tag{12}$$

The subsequent integrals are self-explanatory as μ and σ are simply constants. Integrating them yields:

$$\int_0^t \left(\mu - \frac{1}{2}\sigma^2\right) du = \left(\mu - \frac{1}{2}\sigma^2\right) t \tag{13}$$

and:

$$\int_0^t \sigma dW_u = \sigma W_t \tag{14}$$

Putting this back together gives us then:

$$\ln\left(\frac{S_t}{S_0}\right) = \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t \tag{15}$$

and exponentiating we get:

$$\frac{S_t}{S_0} = \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right) \tag{16}$$

and moving S_0 finally achieves our solution of:

$$S_t = S_0 \cdot \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right) \quad \text{[QED]} \tag{17}$$

3 Statistical Methodology

Now that we've demonstrated one of the core equations quantitative traders and market fanatics alike rely on to simulate and forecast financial asset prices, it is now time to outline the statistical methodology with which we will test it against real market data, using the S&P 500 as our target index and as a benchmark for broad equity market dynamics. This data will be gathered using a ten year daily time series (September 2016–Sept 2026) so that we may capture a diverse array of market regimes, such as bull runs, drawdowns, and overall high volatility environments.

In order to accomplish this, we'll employ a Log daily return series (r_t) that is as such:

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right) = \ln(S_t) - \ln(S_{t-1}) \tag{18}$$

This usage of a log return series allows for continuous additivity across all time intervals we analyze, and it further aligns directly with the continuous-time framework present in Section 2.

It is also worth outlining the theoretical assumptions of Geometric Brownian Motion at this time:

1. Gaussian Distribution, i.e., log returns follow a normal distribution as such:

$$r_t \sim \mathcal{N}\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t,\, \sigma^2 \Delta t\right) \tag{19}$$

2. Independence, i.e., increments of dW_t are independent of one another, implying zero-serial autocorrelation over non-overlapping time steps / time increments.

3. Homoscedasticity, i.e., volatility (σ) is constant across all t.

In order to now actually evaluate the question as to whether real market returns adhere to the above assumptions (or the Gaussian hypothesis), we below defined our core statistical metrics:

3.1 Skewness

Skewness (S) is defined as:

$$S = \frac{\frac{1}{N}\sum_{t=1}^{N}(r_t - \bar{r})^3}{\left(\frac{1}{N}\sum_{t=1}^{N}(r_t - \bar{r})^2\right)^{3/2}} \tag{20}$$

with a theoretical normal value of $S = 0$.

3.2 Kurtosis

Kurtosis (K) or tail weight is defined as:

$$K = \frac{\frac{1}{N}\sum_{t=1}^{N}(r_t - \bar{r})^4}{\left(\frac{1}{N}\sum_{t=1}^{N}(r_t - \bar{r})^2\right)^{2}} \tag{21}$$

with a theoretical normal value of $K = 3$.

3.3 Jarque-Bera Goodness-of-Fit Test

(for context: a JB Goodness of fit test is a joint hypothetical test measuring whether sample Skewness & Kurtosis are in agreement with a normal distribution) [JB87]
It is defined as such:

$$JB = \frac{N}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2(2) \tag{22}$$

H_0: Data is normal, i.e., $S = 0$ and $K = 3$
if $p < .05$, H_0 is rejected, proving deviation from GBM.

3.4 Ljung-Box Test (for Autocorrelation)

This test is essentially to evaluate squared returns vs. standard returns [LB78]. However, it is explicitly defined as the following:

3.4.1 Autocorrelation Function (ACF)

For a log-return series r_t, with mean of $\bar{r}$ and a total observations of N, with lag at k, the sample autocorrelation function is:

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{N}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{N}(r_t - \bar{r})^2} \tag{23}$$

- $\hat{\rho}_k$ measures the linear dependence between returns separated by k trading days.
- Theoretical autocorrelation is zero ($\hat{\rho}_k = 0$) for all $k \geq 1$.

3.4.2 Calculating the Ljung-Box Q-Statistic

In order to test independence across a block or group of m lags simultaneously ($m \leq 10$ or 20 or 100 trading days), the LB-Q stat is computed as:

$$Q = N(N+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{N-k} \tag{24}$$

where:
N = total observation
m = number of lags tested
$\frac{N+2}{N-k}$: a finite sample weighting factor that gives greater weight to shorter lags, serving to improve test performance in smaller samples.

3.4.3 Hypothesis Testing / χ^2 Distribution

The test statistic for the experiment we are testing follows a χ^2 distribution w/ m degrees of freedom since under H_0 the time series is independent of lag (m).

Meaning the test statistic appears as such:

$$Q \sim \chi^2(m) \tag{25}$$

And our hypotheses follow as such:
$H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0$ (returns are independent and align w/ Geometric Brownian Motion)
H_A : At least one autocorrelation coefficient $\rho_k \neq 0$, meaning returns exhibit some tendency of dependency
P-Val: If the test statistic exceeds critical value ($Q > \chi^2_{1-\alpha,m}$) or if $p < .05$, H_0 is rejected.

3.4.4 Employment of Dual Testing Framework

In order to systematically and rigorously evaluate the GBM, the L-B test is applied in essentially 2 distinct ways:

1. Raw-Logs Testing (r_t): testing for price changing dependency
2. Squared-Logs Testing (r_t^2): testing for volatility clustering

3.5 Lo-MacKinlay Variance Ratio Test

While a typical variance ratio test is designed to test two independent entities, for our particular experiment we will employ a variation of this test, known as the Lo-MacKinlay Time Series Variance Ratio Test [LM88] in order to analyze overlapping windows of returns, scaling variance, and heteroskedasticity using the Lo-MacKinlay test statistic for changing volatility (Z^*). The complete test is detailed below:

$$X_t - X_{t-1} = \mu + \varepsilon_t \tag{26}$$

where $X_t = \log P_t$ (P_t is asset price at time t), μ = drift parameter, and ε_t = disturbance term.

Over a sample of log returns X_0 to X_n, with n observations, let $1 + m = nk$, where k is lag term. The estimation for the mean for a 1-period return is:

$$\hat{\mu} = \frac{1}{n}(X_n - X_0) \tag{27}$$

3.5.1 Variance Estimators

- 1-period return variance ($\hat{\sigma}^2$):

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(X_t - X_{t-1} - \hat{\mu})^2 \tag{28}$$

- k-period overlapping variance ($\hat{\sigma}_k^2$):

$$\hat{\sigma}_k^2 = \frac{1}{m}\sum_{t=k}^{n}(X_t - X_{t-k} - k\hat{\mu})^2 \tag{29}$$

where $m = k \cdot (n-k+1) \cdot \left(1 - \frac{k}{n}\right)$.

3.5.2 Variance Ratio Statistic

$$VR(k) = \frac{\hat{\sigma}_k^2}{\hat{\sigma}^2} = 1 + 2\sum_{j=1}^{k-1}\left(1 - \frac{j}{k}\right)\hat{\rho}_j \tag{30}$$

where $\hat{\rho}_j$ is the jth sample autocorrelation of 1-period returns. Under a random walk, H_0 holds $VR(k) = 1$.

3.5.3 Test Statistic

- Homoskedasticity (M_1): Assumes i.i.d. error terms with constant variance:

$$Z(k) = \frac{VR(k) - 1}{[\phi(k)]^{1/2}} \sim \mathcal{N}(0,1) \tag{31}$$

- Heteroskedasticity (M_2): Assumes relaxation of i.i.d. assumption to allow for varying variance/volatility:

$$Z^*(k) = \frac{VR(k) - 1}{[\phi^*(k)]^{1/2}} \sim \mathcal{N}(0,1) \tag{32}$$

where $\phi^*(k)$ incorporates time-variance 4th moments.

3.6 Augmented Dickey-Fuller Test (ADF)

This test serves to demonstrate if a time series is stationary or contains a unit root, meaning a non-stationary random walk.[2] While the Lo-MacKinlay Variance Ratio test evaluates if variance scales linearly or over a certain lag (k), this test, outlined below, observes autoregressive decay parameters.

3.6.1 Model

$X_t = \log P_t$, the standardized $AR(p)$ specification w/ drift α & linear trend βt is:

$$X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{i=1}^{p-1}\phi_i \Delta X_{t-i} + \varepsilon_t \tag{33}$$

In standard ADF regression form, after subtracting X_{t-1} from both sides:

$$\Delta X_t = \alpha + \beta t + \pi X_{t-1} + \sum_{i=1}^{p-1}\psi_i \Delta X_{t-i} + \varepsilon_t \tag{34}$$

where:

$$\begin{aligned}
\Delta X_t &= X_t - X_{t-1} \\
\pi &= \gamma - 1 \\
\psi_i &= -\sum_{j=i+1}^{p}\phi_j \\
\varepsilon_t &\sim \text{i.i.d. } (0, \sigma^2) \text{ serves as a disturbance term}
\end{aligned}$$

[2] The unit-root test is due to [DF79]; the augmented version is due to [SD84].

3.6.2 Framework & Hypotheses

$$H_0 : \pi = 0 \quad \text{(unit root present; random walk)}$$
$$H_1 : \pi < 0 \quad \text{(no unit root; mean-reverting)}$$

Test statistic:

$$\text{ADF}_{\text{stat}} = \frac{\hat{\pi}}{\text{SE}(\hat{\pi})} \tag{35}$$

Note that since X_t is non-stationary under H_0, the test statistic for the ADF does not follow a Student's t-distribution, and instead requires critical values derived from asymptotic simulation.

3.7 Ornstein-Uhlenbeck (OU) & AR(1) Mean-Reversion Analysis

The OU process [UO30] serves to quantify the exact velocity of mean reversion & its subsequent statistical half-life.

3.7.1 Stochastic Differential Equation for OU Process

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t \tag{36}$$

where:

$$\theta > 0 : \text{rate of mean reversion}$$
$$\mu : \text{the long run equilibrium mean}$$
$$\sigma > 0 : \text{the continuous volatility parameter}$$
$$W_t : \text{our standard Wiener process}$$

3.7.2 Discrete-Time (AR(1)) Mapping

Integrating the SDE over a discrete time step $\Delta t = 1$ is as follows:

$$dX_t + \theta X_t dt = \theta\mu dt + \sigma dW_t \tag{37}$$

$$e^{\theta t}dX_t + \theta e^{\theta t}X_t dt = \theta\mu e^{\theta t}dt + \sigma e^{\theta t}dW_t \tag{38}$$

Apply Itô's Lemma (defined section 2) to the product $d(e^{\theta t}X_t)$:

$$d(e^{\theta t}X_t) = e^{\theta t}dX_t + \theta e^{\theta t}X_t dt \tag{39}$$

$$d(e^{\theta t}X_t) = \theta\mu e^{\theta t}dt + \sigma e^{\theta t}dW_t \tag{40}$$

Integrate across $[s, t]$ time:

$$\int_s^t d(e^{\theta u}X_u) = \theta\mu \int_s^t e^{\theta u}du + \sigma \int_s^t e^{\theta u}dW_u \tag{41}$$

Evaluating:

$$e^{\theta t}X_t - e^{\theta s}X_s = \mu(e^{\theta t} - e^{\theta s}) + \sigma \int_s^t e^{\theta u}dW_u \tag{42}$$

Isolating X_t & setting $s = t - 1$ gives the one period solution:

$$X_t = \mu(1 - e^{-\theta}) + e^{-\theta}X_{t-1} + \sigma \int_{t-1}^t e^{-\theta(t-u)}dW_u \tag{43}$$

3.7.3 Respective Mean & Variance

Since the integral term above is an Itô integral with a deterministic integrand, ε_t (disturbance term) $\sim \mathcal{N}(0, \cdot)$.

It follows then:

$$\mathbb{E}[\varepsilon_t] = 0 \longrightarrow \mathbb{E}[X_t \mid X_{t-1}] = \mu(1 - e^{-\theta}) + e^{-\theta} X_{t-1} \tag{44}$$

To compute the residual variance:

$$\mathrm{Var}(\varepsilon_t) = \mathrm{Var}\left(\sigma \int_{t-1}^{t} e^{-\theta(t-u)} dW_u\right) \tag{45}$$

$$\mathrm{Var}(\varepsilon_t) = \sigma^2 \int_{t-1}^{t} e^{-2\theta(t-u)} du \tag{46}$$

Substituting $v = t - u \longrightarrow du = -dv$:

$$\mathrm{Var}(\varepsilon_t) = \sigma^2 \int_{0}^{1} e^{-2\theta v} dv \tag{47}$$

Computing that out it follows:

$$\sigma_\varepsilon^2 = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta}) \tag{48}$$

3.7.4 Discrete AR(1) Parameter Identification

Setting $a = \mu(1 - e^{-\theta})$ & $b = e^{-\theta}$ yields:

$$X_t = a + bX_{t-1} + \varepsilon_t \quad ; \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{49}$$

This allows you to recover the continuous time parameters:

$$\theta = -\ln(b) \quad ; \quad \mu = \frac{a}{1-b} \quad ; \quad \sigma = \sigma_\varepsilon \sqrt{\frac{2\theta}{1 - e^{-2\theta}}} \tag{50}$$

3.7.5 Mean Reversion Half-Life

$t_{1/2}$ serves to define the time expected for the process to decay halfway back to its normal long-run mean μ following a shock.

Using the expected integrated SDE on condition X_0‘:

$$\mathbb{E}[X_t \mid X_0] - \mu = (X_0 - \mu)e^{-\theta t} \tag{51}$$

Setting:

$$\frac{\mathbb{E}[X_{t+t_{1/2}} \mid X_0] - \mu}{X_0 - \mu} = \frac{1}{2} \tag{52}$$

We find:

$$t_{1/2} = \frac{\ln(2)}{-\ln(b)} \tag{53}$$

4 Empirical Results and Analysis

The findings discussed in this section will be strictly concerned with the experimental data collected to fulfill our statistical frameworks formerly outlined in Section 3. The results are reviewed below:

Table 1: Distributional Moments and Normality Test

Metric / Test	**Obs (N)**	**Sample Value**	**GBM Target**	**Test Stat (JB)**	**p-value**
Skewness	2,519	-0.6776	0.0000	—	—
Kurtosis	2,519	19.6907	3.0000	—	—
Jarque-Bera Test	2,519	—	Normality	29,431.83	< 0.0001

4.1 Jarque-Bera Goodness of Fit Test

The skewness value -0.6776 (leftward skew) implies that market drawdowns happen more rapidly than market growth, contradicting the premise of normally distributed returns. The Kurtosis value of 19.6907 demonstrates leptokurtosis. This occurred for several reasons:

1. Changing Conditions: Markets experience longer periods of lower volatility and shorter bursts of higher volatility. This also serves as a reason for the leftward skew mentioned above.

2. Lack of Independence: Market systems operate under multi-dependent frameworks, meaning that selloffs often drive other selloffs.

3. Mean Reversion: Markets are mean-reverting systems, and this concentration around the mean, combined with infrequent large-scale deviations, drives both uni-modal peaks and fat tailing.

4.2 Ljung-Box Test

Table 2: Ljung-Box Test on Raw Log-Returns (Q_{raw}) and Squared Returns (Q_{sq})

Lag (k)	**Q_{raw} Stat**	**p-value (Raw)**	**Q_{sq} Stat**	**p-value (Sq)**
10	255.6934	< 0.0001	2,741.2237	< 0.0001
20	307.0394	< 0.0001	3,173.1028	< 0.0001
100	427.1381	< 0.0001	3,281.7427	< 0.0001

This illustrates:

1. $Q \to p$**:** Strong linear dependency is illustrated, as is mean-reversion, holding that future returns are directly related to past returns, making a rand walk modeling system less statistically sound for model data that generates such raw returns.

2. $Q^2 \to p^2$**:** The magnitude of the squared returns illustrates strong volatility clustering, giving way to possible Autoregressive Conditional Heteroskedasticity.

4.3 Lo-MacKinlay Variance Ratio Test

Table 3: Variance Ratio Test Results

Series	q	VR	z_H	p_H	z_{He}	p_{He}	Type
Log-price	2	0.856	-7.233	4.73×10^{-13}	-2.355	0.0185	Price
Log-price	5	0.838	-3.721	0.000199	-1.227	0.2198	Price
Log-price	10	0.811	-2.805	0.00503	-0.975	0.3298	Price
Log-price	20	0.806	-1.959	0.0501	-0.736	0.4615	Price
Log-price	60	0.624	-2.136	0.0326	-1.012	0.3114	Price
Realized vol.	2	1.209	10.434	0	4.008	6.12×10^{-5}	Volatility
Realized vol.	5	1.806	18.393	0	7.300	2.88×10^{-13}	Volatility
Realized vol.	10	2.579	23.375	0	9.859	0	Volatility
Realized vol.	2	1.328	16.265	0	4.131	3.61×10^{-5}	Volatility
Realized vol.	5	2.239	28.016	0	7.344	2.07×10^{-13}	Volatility
Realized vol.	10	3.509	36.827	0	10.371	0	Volatility
Realized vol.	21	5.458	43.300	0	14.114	0	Volatility

On the log-price, $VR(q)$ sits below 1 at every horizon, which at first glance suggests mean reversion. However, only $q = 2$ survives the heteroskedasticity-robust statistic ($z^* = -2.36$, $p = 0.019$); for $q \geq 5$ the robust p-values range from 0.22 to 0.46. The apparent reversal is therefore driven by a handful of violent, high-volatility episodes rather than a steady tendency for prices to revert. Realized volatility tells the opposite story: $VR(q)$ rises well above 1 and grows with q (up to 5.46 at $q = 21$ for the 63-day window), meaning volatility shocks build on one another rather than fading.

4.4 Augmented Dickey Fuller Test

Table 4: Augmented Dickey–Fuller Test Results

Series	n	Lag	ADF Stat.	p-value	1%	5%	Result
	2510	9	-0.480	0.896	-3.433	-2.863	Fail to reject
Realized Vol. (21d)	2471	27	-5.082	1.52×10^{-5}	-3.433	-2.863	Reject
Realized Vol. (63d)	2446	10	-4.468	0.000224	-3.433	-2.863	Reject

The log-price fails to reject a unit root ($p = 0.896$), consistent with GBM's random walk. Both realized volatility series reject it ($p < 0.001$), indicating that volatility is stationary and mean-reverting. We note, however, that a rolling-window volatility series will reject a unit root even under constant volatility, since overlapping windows smooth the series; this result alone does not prove GBM wrong, and Section 6 returns to it by simulation.

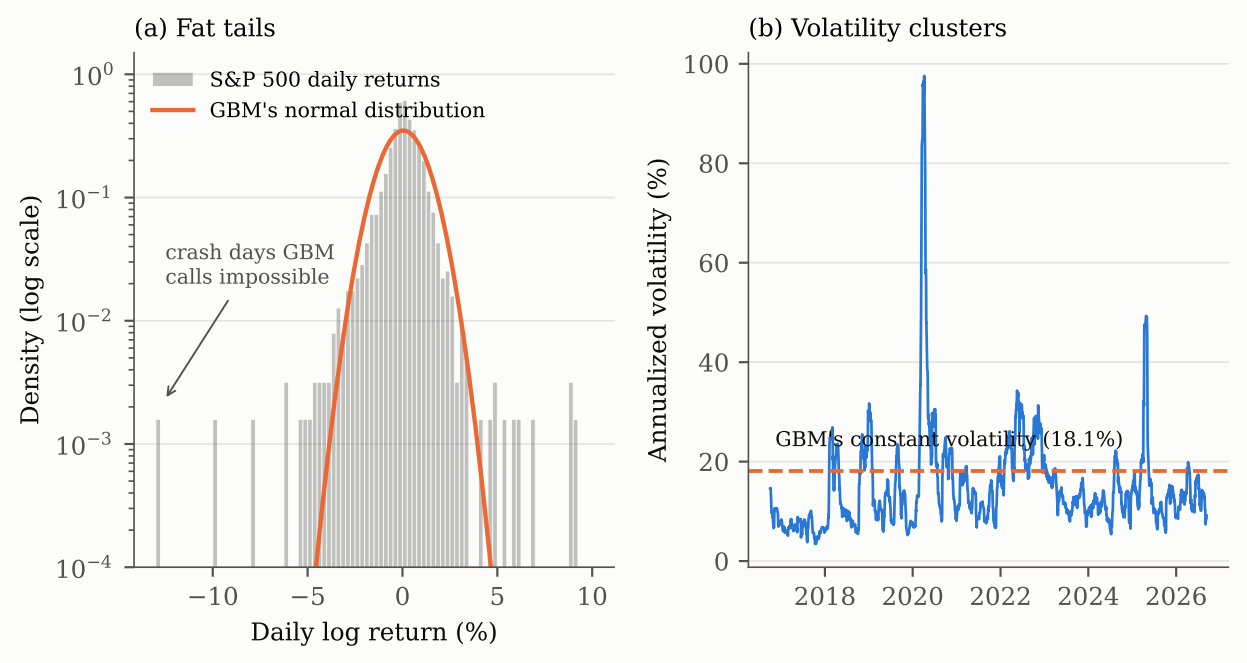
4.5 Ornstein-Uhlenbeck (OU) & AR(1) Mean-Reversion Analysis

Table 5: Regression and Mean Reversion Analysis Results

n	a	b	SE_{HAC}	p	R^2	κ	θ	HL (yr)	HL (trading days)
2498	0.001010	0.993123	0.013882	0	0.986168	1.738904	0.146882	0.398611	100.45
2456	0.000258	0.998382	0.006074	0	0.996798	0.408151	0.159695	1.698260	427.96

It follows then that volatility reverts, but slowly[3]: the half-life of a volatility shock is 100.5 trading days for the 21-day window and 428 days for the 63-day window. Since overlapping windows alone produce half-lives of only about 13 and 41 days under constant volatility, the excess persistence is genuine. GBM, which holds σ fixed, has no mechanism to produce it.

[3]The standard errors reported in Table 5 (SE_{HAC}) are heteroskedasticity- and autocorrelation-consistent in the sense of [NW87].



References

[Bac00] L. Bachelier, *Théorie de la spéculation*, Ann. Sci. École Norm. Sup. (3) **17** (1900), 21–86.

[BS73] F. Black and M. Scholes, *The pricing of options and corporate liabilities*, J. Polit. Econ. **81** (1973), no. 3, 637–654.

[DF79] D. A. Dickey and W. A. Fuller, *Distribution of the estimators for autoregressive time series with a unit root*, J. Amer. Statist. Assoc. **74** (1979), no. 366, 427–431.

[JB87] C. M. Jarque and A. K. Bera, *A test for normality of observations and regression residuals*, Internat. Statist. Rev. **55** (1987), no. 2, 163–172.

[LB78] G. M. Ljung and G. E. P. Box, *On a measure of lack of fit in time series models*, Biometrika **65** (1978), no. 2, 297–303.

[LM88] A. W. Lo and A. C. MacKinlay, *Stock market prices do not follow random walks: evidence from a simple specification test*, Rev. Financ. Stud. **1** (1988), no. 1, 41–66.

[Mer73] R. C. Merton, *Theory of rational option pricing*, Bell J. Econom. Management Sci. **4** (1973), no. 1, 141–183.

[NW87] W. K. Newey and K. D. West, *A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix*, Econometrica **55** (1987), no. 3, 703–708.

[Osb59] M. F. M. Osborne, *Brownian motion in the stock market*, Oper. Res. **7** (1959), no. 2, 145–173.

[SD84] S. E. Said and D. A. Dickey, *Testing for unit roots in autoregressive-moving average models of unknown order*, Biometrika **71** (1984), no. 3, 599–607.

[UO30] G. E. Uhlenbeck and L. S. Ornstein, *On the theory of the Brownian motion*, Phys. Rev. **36** (1930), no. 5, 823–841.